

Annexure-B

BCA

Fifth and Sixth Semesters Syllabus, 2026-27

University of Mysore

Curriculum
Program: BCA
Subject: Computer Applications

Semester	Course No.	Theory/ Practical	Credits	L-T-P	No. of Hours	Paper Title	Marks	
							SEE	CIE
V	CAM51T	Theory	03	3-0-0	03	Network Security	80	20
	CAM51P	Practical	02	0-0-2	04	Network Security Lab	40	10
	CAM52T	Theory	03	3-0-0	03	Design and Analysis of Algorithms	80	20
	CAM52P	Practical	02	0-0-2	04	Design and Analysis of Algorithms Lab	40	10
	CAM53T	Theory	03	3-0-0	03	Computer Graphics & Multimedia	80	20
	CAM53P	Practical	02	0-0-2	04	Computer Graphics & Multimedia Lab	40	10
	CAM54T	Theory	03	3-0-0	03	Blockchain Technology	80	20
	CAM55T	Theory	03	3-0-0	03	Enterprise Resource Planning	80	20
	CAM56T	Theory	03	3-0-0	03	Fundamentals of Artificial Intelligence	80	20
	CACP51P	Practical (Compulsory Paper)	02	0-0-2	04	AI Tools and its applications	40	10

Semester: V

Course Code:	Course Title: Network Security
Course Credits: 03 (3:0:0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Build foundational knowledge.

CLO 2: Focus on cryptography and secure communication.

CLO 3: Emphasize practical tools and mechanisms.

CLO 4: Introduce advanced, real-world applications and legal aspects.

Course Pedagogy:

1. Interactive Lectures and Conceptual Discussions to introduce network security principles, cryptographic techniques, and security protocols.
2. Case Studies and Real-world Security Incidents Analysis to understand cyber-attacks, security breaches, and preventive measures.
3. Student Seminars, Assignments focusing on emerging trends such as cloud security, IoT security, ethical hacking, and cyber laws.

Course Content	Hours
Unit 1:	
Fundamentals of Network Security - Introduction to Information Security: CIA triad (Confidentiality, Integrity, Availability), Threats and Attacks: Malware, phishing, denial of service, man-in-the-middle, Security Services: Authentication, authorization, non-repudiation, Cryptography Basics: Symmetric vs. asymmetric encryption, hashing, digital signatures, Overview of Security Policies and Standards (ISO/IEC 27001, NIST).	11
Unit 2:	
Cryptography and Secure Communication - Symmetric Key Algorithms: DES, AES, Asymmetric Key Algorithms: RSA, Diffie-Hellman key exchange, Hash Functions: MD5, SHA family, Public Key Infrastructure (PKI) and Certificates, Secure Communication Protocols: SSL/TLS, HTTPS, IPSec, VPNs, Email Security: PGP, S/MIME.	11
Unit 3:	

Network Security Tools and Mechanisms - Firewalls: Types (packet filtering, proxy, stateful inspection), configuration basics, Intrusion Detection and Prevention Systems (IDS/IPS), Virtual Private Networks (VPNs) and tunneling protocols, Wireless Security: WEP, WPA, WPA2, WPA3, Security in Cloud and IoT environments, Case Studies: Real-world breaches and lessons learned.	11
Unit 4:	
Advanced Topics and Management - Security in Operating Systems and Applications, Incident Response and Disaster Recovery Planning, Security Auditing and Vulnerability Assessment, Ethical Hacking and Penetration Testing basics, Emerging Trends: Zero Trust Architecture, AI in Security, Blockchain for Security, Legal and Ethical Issues: Cyber laws in India, IT Act 2000, GDPR basics.	11

Learning Resources:

1. Atul Kahate - Cryptography and Network Security (Tata McGraw Hill)
2. Data Communications and Networking– Behrouz A. Forouzan (adapted Indian edition, McGraw Hill)
3. William Stallings – Cryptography and Network Security: Principles and Practice (Pearson)
4. William Stallings – Network Security Essentials: Applications and Standards (Pearson)
5. Saurabh Sharma – Information Security and Cyber Laws (Katson Books)
6. Charlie Kaufman, Radia Perlman, Mike Speciner – Network Security: Private Communication in a Public World (Pearson)

Semester: V

Course Code:	Course Title: Network Security Lab
Course Credits: 02 (0:0:2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Demonstrate the use of network security tools such as Wireshark and Nmap for monitoring and analyzing network traffic.

CLO 2: Implement basic cryptographic algorithms such as Caesar Cipher, Playfair Cipher, Hill Cipher, and RSA for secure communication.

CLO 3: Apply network security mechanisms including firewall configuration

Course Pedagogy:

1. Hands-on Laboratory Sessions where students practice network monitoring, encryption techniques, and security tool usage.
2. Demonstration-based Teaching by the instructor to explain the working of network security tools and cryptographic algorithms.
3. Problem-solving and Programming Exercises for implementing encryption, decryption, and security algorithms.

List of Practical Experiments:**Part -A**

1. Program to implement the Caesar Cipher substitution method.
2. Program to implement Mono alphabetic Cipher with the given key and input.
3. Program to implement Poly alphabetic Cipher
4. Program to implement the Playfair Cipher algorithm for secure message encryption.
5. Program to implement the Hill Cipher algorithm for encryption and decryption.
6. Program to implement single and double transposition techniques

Part - B

7. Program to implement simple DES algorithm
8. Demonstration of symmetric key encryption using AES algorithm.
9. Program to implement RSA algorithm for public key encryption and decryption.
10. Program to generate and verify Digital Signatures using cryptographic techniques.
11. Generate Pseudo Random Numbers using Linear Congruential method
12. Demonstrate Diffie Helmn Key Exchange Algorithm.

Evaluation Scheme for Practical Examination:

Assessment Criteria		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written programs	5
Viva Voce		5
Total		40

Semester: V

Course code:	Course title: Design and Analysis of Algorithms
Course credits:03 (3:0:0)	Hours/Week: 3
Total contact class: 44	Formative assessment marks:20
Exam marks: 80	Exam duration: 3 Hours

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

- CLO 1:** Understand fundamental concepts of algorithms and analyze their time and space complexity using asymptotic notations.
- CLO 2:** Apply brute force and divide-and-conquer techniques to solve searching and sorting problems.
- CLO 3:** Design solutions for optimization problems using greedy method and understand minimum spanning tree algorithms.
- CLO 4:** Know the concepts of backtracking and dynamic programming and apply them to solve computational problems.

Course pedagogy

1. Interactive classroom teaching using board and PPT to explain algorithm design techniques, asymptotic analysis, and problem-solving strategies.
2. Step-by-step demonstration of algorithms like Binary Search, Quick Sort, Merge Sort, Prim's and Kruskal's algorithms.
3. Practical/Programming Approach to implement algorithms using C/Python to understand efficiency and complexity analysis.
4. Problem Solving approach to solve numerical problems on time complexity, Greedy method, Knapsack, N-Queens and Travelling Salesman Problem.

- Interactive Learning Methods: Assignments, quizzes, group discussions and case studies to improve analytical and logical thinking skills.

Course contents	Hours
Unit 1:	
INTRODUCTION: Algorithm, Pseudo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, Worst-case, Best-case and Average-case efficiencies, Asymptotic Notation, Big oh notation, Omega notation, Theta notation. Mathematical analysis of non-recursive algorithms, mathematical analysis of recursive algorithms.	11
Unit 2:	
Brute Force and Exhaustive Search: Introduction to Brute Force approach, Selection Sort and Bubble Sort, Sequential search, Exhaustive Search- Travelling Salesman Problem and Knapsack Problem. Divide and Conquer:	11
General method, Applications-Binary search, Quick sort, Merge sort	
Unit 3:	
Greedy Method- Introduction to Greedy Method, Definition of Greedy algorithm, Characteristics of Greedy Strategy, General structure of greedy algorithms, Fractional Knapsack Problem, Job Sequencing with Deadlines, Minimum Spanning Tree- Prim's Algorithm, Kruskal's Algorithm.	11
Unit 4:	
Transform and Conquer: Introduction, Heaps and Heap Sort Backtracking: Introduction, N-Queens problem (4 queens), Introduction to Dynamic Programming, need of dynamic programming, introduction to dynamic programming approaches	11

Learning Resources:

- Introduction to the Design and Analysis of Algorithms, Anany Levitin: 2nd Edition, 2009, Pearson.
- Ellis Horowitz, Satraj Sahni and Rajasekharan, Fundamentals of Computer Algorithms, 2nd Edition, University Press, 2008
- Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
- Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)
- Design and Analysis of Algorithms, P. H. Dave, H. B. Dave, 2nd edition, Pearson Education

Semester: V

Course code:	Course title: Design and Analysis of Algorithms Lab
Course credits:02 (0:0:2)	Hours/Week: 4
Total contact class: 60	Formative assessment marks: 10
Exam marks: 40	Exam duration: 3 Hours

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

CLO 1: Implement searching and sorting algorithms.

CLO 2: Analyze time and space complexity of algorithms.

CLO 3: Implement divide and conquer and greedy algorithms.

CLO 4: Apply algorithm design techniques to solve computational problems.

Course Pedagogy:

1. Design algorithm based various techniques.
2. Implementation of various sorting and searching algorithms.
3. Hands-on experiments to implement algorithms and analyze complexity.
4. Students plot graph to compare apriori and aposteriori analysis.

List of practical experiments

1. Program to search an element by sequential search method and estimate the time and space complexity
2. Program to generate Fibonacci series without recursion and estimate the time and space complexity
3. Program to search an element by binary search and estimate the time and space Complexity
4. Program to find factorial of a number using recursion and estimate the time and space complexity
5. Program to implement merge sort algorithm and estimate the time and space complexity
6. Program to generate N random numbers and sort them using quick sort.
7. Write a program to find minimum and maximum value in an array using divide and Conquer
8. Implementation of Fractional Knapsack Algorithm using Greedy Approach
9. Program to find the minimum cost spanning tree using prim's algorithm and estimate time space complexity
10. Program to find the minimum cost spanning tree using Kruskal's algorithm and estimate the time and space complexity

General instructions

1. Design the Algorithm
2. Perform apriori analysis and plot the graph
3. Implement the algorithm in C++
4. Tabulate the Results for different set of inputs
5. Perform aposteriori analysis and plot the Graph

Evaluation scheme for practical examination:

Assessment Criteria		Marks
Writing Only (One problem)	Writing the Algorithm and apriori analysis	10
	Writing the given Program	15
Execution	Execution of the program	5
Result Analysis using aposteriori analysis by plotting Graph	Tabulating the results for different set of inputs	5
Viva Voce		5
Total		40

Semester: V

Course Code:	Course Title: Computer Graphics & Multimedia
Course Credits: 03 (3:0:0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Understand the fundamentals of computer graphics and multimedia

CLO 2: Apply different graphics algorithms for drawing and displaying graphical objects such as lines, circles and polygons.

CLO3: Implement geometric transformations such as translation, rotation, scaling, reflection, and shearing in computer graphic

CLO4: Analyze multimedia elements such as text, images, audio, video, and animation used in multimedia applications.

CLO5: Use multimedia tools and software such as Adobe Photoshop, Blender, and Adobe Animate for creating multimedia content.

Course Pedagogy:

1. Classroom lectures to explain fundamental concepts of computer graphics and multimedia systems.
2. Topics include graphics algorithms, transformations, and multimedia components 3. Demonstration of graphics techniques such as line drawing, clipping, and transformations.
4. Students practice drawing objects, applying transformations, and creating animations.

Course Content	Hours
Unit 1:	
Computer graphics fundamentals – Introduction, Application Areas of Computer Graphics, Overview of Graphics Systems, Video-Display Devices, Raster-Scan Systems, Random Scan Systems, Graphics Monitors and Work Stations and Input Devices. Points and Lines - Line Drawing Algorithms, Mid-Point Circle and Ellipse Algorithms, Scan Line Polygon Fill Algorithm, Boundary Fill and Flood-Fill Algorithms. Curve Representation: Parametric Curves, Parametric Representation of a Circle, Parametric representation of cubic curves, drawing Bezier curve	11
Unit 2:	

Two-Dimensional Transformations: Coordinate and Geometric Transformations; Translation, Rotation, Scaling; Matrix representations and Homogeneous coordinates, Composite transformations, General Pivot Point rotation, General Fixed-Point Scaling, Shearing; Reflection; Reflection about an arbitrary line. 2-D Viewing: Viewing pipeline; Window, Viewport, Window-to-Viewport transformation; Zooming, panning; Pointing and Positioning techniques; Rubber band technique; Dragging. Clipping	11
operations: Point and Line clipping, Cohen-Sutherland line clipping, Mid-Point Subdivision line clipping.	
Unit 3:	
3-D Graphics & Modeling: Visualization techniques for Realism; 3D Object Representation; Solid Model Representation Schemes; Euclidean Geometry methods: Regularized Boolean Set Operations, Primitive Instancing, Boundary Representations, Curved lines and surfaces, Sweep Representations, Spatial-Partitioning Representations - Octree representation, Constructive Solid Geometry; Procedural Methods: Fractals, Shape Grammars, Particle systems, Physically Based modeling, Visualization techniques; 3D transformations. Three-Dimensional Viewing: Viewing Pipeline; Parallel Projection: Orthographic and Oblique Projection.	11
Unit 4:	
Introduction to Animation: Fundamentals of Animation, Traditional Animation, Computer Animation, Types of computer Animation, Challenges and the future of computer animation Designing of Animation Sequences; Key-Frame Systems; Introduction to Photoshop: Drawing and Creating Design, Designing Pamphlets, Image Color Corrections, Morphing One Image with the Other Images, Doing B&W Photo to Color Designing & Crating Logos Working with Brush, Pen, Text Tools Blending Options Creating Text Effects Designing Layout for the Web Page Designing Web Banners and Buttons Doing Actions in Photoshop Doing Animation in Photoshop Creating Posters with Text Effects Creating the Out Line Designs Drawing the Basic Shapes with Pencil and Pen Designing the Frame Layouts for Photos.	11

Learning Resources:

1. Hearn D and Baker P. "Computer Graphics Open GL Version", Fourth Edition, Prentice Hall of India; 2013.
2. Lance Flavell, "Beginning Blender: Open-source 3D Modeling, Animation, and Game Design", Apress, 2010
3. James Chronister, "Blender basics classroom tutorial book", Fourth edition, 2011
4. Plastook R A and Kalley G. "Theory and Problems of Computer Graphics", Schaum's Outline Series, TMH;1985.
5. Foley J D D, Eiener S K and Hughes J.F. "Computer Graphics Principles and Practice", Second Edition, Pearson Education;1996.

Course Code:	Course Title: Computer Graphics & Multimedia Lab
Course Credits: 02 (0:0:2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

- CLO 1:** Implement basic computer graphics algorithms to draw lines, circles, and other graphical objects.
- CLO 2:** Apply 2D transformations such as translation, rotation, scaling, reflection, and shearing to graphical object
- CLO 3:** Implement clipping and filling algorithms used in computer graphics application.
- CLO 4:** Create simple graphics programs to display shapes, patterns, and animations.
- CLO 5:** Use multimedia tools such as Adobe Photoshop, Audacity, and Blender for editing images, audio, and animation.

Course pedagogy:

1. Demonstrates graphics algorithms, image editing, audio/video editing, and animation techniques before the students perform the experiments.
2. Hands-on Practical Sessions: Students perform lab experiments to implement graphics algorithms and create multimedia content using tools such as Adobe Photoshop, Audacity and Blender.

List of Practical Experiments:

Part A

1. Program to draw basic graphics primitives
2. Program to implement Digital Differential Analyzer (DDA) Line Drawing Algorithm.
3. Program to implement Bresenham's Line Drawing Algorithm
4. Program to draw a circle using Bresenham's Circle Algorithm
5. Program to perform 2D Transformations

6. Program for Line Clipping using Cohen–Sutherland Algorithm
7. Program for Polygon Clipping
8. Program to perform 3D Transformations

(Note: The above-mentioned programs are implemented using C++)

Part B

1. Create and edit images using image editing software such as Adobe Photoshop or GIMP.
2. Perform image enhancement operations such as cropping, resizing, brightness, and contrast adjustment.
3. Design a poster or banner using graphics tools.
4. Record and edit audio using software such as Audacity.
5. Perform audio operations such as noise reduction, trimming, mixing, and adding sound effects.
6. Create a short audio clip with background music and voice narration.
7. Create and edit videos using tools such as Adobe Premiere Pro or OpenShot Video Editor.
8. Create a short video presentation or advertisement.

Evaluation Scheme for Lab Examination:

Assessment Criteria		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written programs	5
Viva Voce		5
Total		40

Course Code:	Course Title: Blockchain Technology
Course Credits: 03 (3:0:0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

- CLO 1:** Describe the fundamental concepts of blockchain technology including distributed ledger systems, block structure, hashing, and types of blockchain.
- CLO 2:** Demonstrate the role of cryptographic techniques such as hash functions, digital signatures, and public key cryptography in ensuring blockchain security.
- CLO 3:** Analyze the working of cryptocurrencies and Bitcoin including transaction processes, mining, and consensus mechanisms.
- CLO 4:** Apply the concepts of smart contracts and blockchain platforms such as Ethereum and Hyperledger to understand real-world blockchain applications.

Course Pedagogy

1. Interactive Lectures and Conceptual Discussions to explain blockchain architecture, distributed systems, and cryptographic principles.
2. Hands-on Sessions on blockchain tools, cryptocurrency transactions, and smart contract platforms.
3. Case Studies and Real-world Examples focusing on blockchain applications in finance, supply chain management, healthcare, and governance.
4. Student Seminars / Mini Projects where students explore emerging blockchain platforms, cryptocurrencies, or smart contract applications.

Content	Hours
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Unit 1:	
Introduction to Blockchain Technology - Evolution of Digital Systems, Introduction to Blockchain Technology, Distributed Systems and Distributed Ledger Technology (DLT), Blockchain Architecture and Components, Structure of Blocks, Hashing Concept, Chain Formation Process, Centralized vs Decentralized Systems, Types of Blockchains, Advantages of Blockchain, Limitations and Challenges	11
Unit 2:	
Cryptography in Blockchain - Introduction to Cryptography, Cryptographic Hash Functions, Digital Signatures, Public Key and Private Key Cryptography, Merkle Trees, Role of Cryptography in Blockchain, Wallets and Blockchain Addresses, Blockchain Security Concepts, Double Spending Problem, 51% Attack, Privacy and Security Issues.	11
Unit 3:	
Cryptocurrencies and Bitcoin - Introduction to Cryptocurrencies, History and Evolution of Bitcoin, Bitcoin Network Architecture, Bitcoin Transactions, Blockchain Transaction Lifecycle, Mining Process, Proof of Work Consensus Mechanism, Cryptocurrency Wallets, Cryptocurrency Exchanges, Advantages and Limitations of Bitcoin and Regulatory Issues in Cryptocurrency.	11
Unit 4:	
Smart Contracts and Blockchain Platforms - Introduction to Smart Contracts, Working of Smart Contracts, Features and Benefits of Smart Contracts, Introduction to Ethereum, Ethereum Architecture, Ethereum Virtual Machine (EVM), Gas and Transaction Fees, Blockchain Platforms Overview, Comparison of Ethereum, Hyperledger, and Binance Smart Chain, Use Cases of Smart Contracts.	11

Learning Resources:

1. Blockchain Basics: A Non-Technical Introduction in 25 Steps – Daniel Drescher, Apress Publications.
2. Mastering Bitcoin – Andreas M. Antonopoulos, O'Reilly Media.
3. Mastering Ethereum – Andreas M. Antonopoulos and Gavin Wood, O'Reilly Media.
4. Blockchain Technology Explained – Alan T. Norman, CreateSpace Independent Publishing.
5. Blockchain Revolution – Don Tapscott and Alex Tapscott, Portfolio Publications.
6. Introducing Blockchain with Lisp – Ken Huang, Apress Publications.
7. Blockchain: Blueprint for a New Economy – Melanie Swan, O'Reilly Media.

Semester: V

Course code:	Course title: Enterprise Resource Planning
Course credits:03(3:0:0)	Hours/Week: 3
Total contact class: 44	Formative assessment marks :20
Exam marks: 80	Exam duration: 3 Hours

Course Learning Outcomes:

Upon successful completion of this course, students will be able to, **CLO1:**
Understand ERP Fundamentals.

CLO 2: Integrate Related Technologies. **CLO**

3: Evaluate ERP Marketplace

CLO 4: Manage Implementation Lifecycle.

CLO 5: Adapt to Emerging Trends.

Course pedagogy

1. Interactive Lectures and Conceptual Discussions
2. Hands-on Demonstrations and Tool-based Learning.
3. Case Studies and Industry-oriented Analysis
4. Assignments, Student Seminars/Presentations

Course Contents	Hours
Unit 1: Enterprise Resource Planning (ERP): Introduction, history, advantages Enterprise: Introduction, business modeling, integrated data model, integrated management information, Basic concepts of ERP, Risks and benefits of ERP	11
Unit 2: ERP and Related Technologies: Introduction to MRP, MRP-II and ERP Business Process Reengineering, (BPR), Data warehousing, data mining and Online Analytical Processing (OLAP), Product Life Cycle Management (PLM), Supply Chain Management (SCM), Customer Relationship Management (CRM)	11
Unit 3: ERP Marketplace: overview, dynamics, ERP- the Indian Scenario, Functional modules of ERP software, Integration of ERP, SCM and CRM.	11
Unit 4: ERP package selection, ERP Implementation basics, ERP Implementation Life Cycle. ERP Maintenance and Support, Data Migration and Performance Measurement, Maximizing and Scaling up ERP Systems, Emerging Trends-Cloud ERP, Mobile ERP.	11

Semester: V

Learning Resources:

1. ERP Demystified, Alexis Leon, 3rd Edition, McGraw-Hill Education, 2014.
2. Concepts in Enterprise Resource Planning, Joseph A. Brady, Ellen F. Monk, Bret J. Wagner, Cengage Learning, 2001.
3. Enterprise Resource Planning – Concepts and Practice, Vinod Kumar Garg, N.K. Venkitakrishnan, 2nd Edition, Prentice-Hall of India Pvt. Ltd., 2008.

Course code:	Course title: Fundamentals of Artificial Intelligence
Course credits:03 (3:0:0)	Hours/Week: 3
Total contact class: 44	Formative assessment marks:20
Exam marks: 80	Exam duration: 3 Hours

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

CLO 1: Explain the basic concepts, history, scope, and applications of Artificial Intelligence

CLO 2: Describe intelligent agents, their structure, types, and apply basic AI problem- solving and search techniques.

CLO 3: Understand Machine Learning fundamentals including supervised, unsupervised and Reinforcement learning concepts.

CLO 4: Gain the knowledge of representation techniques, logic-based representation and Expert systems.

CLO 5: Demonstrate basic usage of AI tools for simple real-world applications.

Course Pedagogy

1. Interactive Lecture Method in Classroom teaching using board and PPT for AI concepts.
2. Hands-on exercises using Python and AI tools.
3. Demonstrating AI applications and tools.
4. Group discussions, seminars, quizzes, and problem-solving tasks to improve understanding and analytical skills.

Course contents	Hours
Unit 1:	
Definition, Scope, and Importance of AI, History and Evolution of AI, Overview of AI system, AI Applications in Real-world: Healthcare, Education, Banking, etc. AI vs. Human Intelligence, Types of AI: Narrow AI, General AI. Ethics and Challenges in AI	11
Unit 2:	
Intelligent agents-definition and structure of intelligent agents, Types-simple, reflex, model based, goal based, utility based. Problem-Solving Techniques in AI, problem formulation, State Space Representation, Search techniques- Breadth-First Search, Depth-First Search, Informed Search: Best-First Search, A* Algorithm, Heuristics in AI, Constraint Satisfaction Problems	11
Unit 3:	

Semester: V

Machine Learning Overview, Types- Supervised, Unsupervised, reinforcement learning, Machine Learning concept- training, testing, validation, classification and clustering, Knowledge Representation in AI - Logic-Based Representation: Propositional and Predicate Logic, Rule-Based Systems and Expert System	11
Unit 4:	
Deep Learning: Concept of Deep Learning, Introduction to Neural Networks, Types of Deep Learning models, Deep learning applications, AI Tools introduction: ChatGPT, DALL-E, Canva AI, Teachable Machine.	11

Learning Resources:

1. Artificial Intelligence: A Modern Approach – Stuart Russell and Peter Norvig, Pearson.
2. Artificial Intelligence – Elaine Rich, Kevin Knight, McGraw Hill.
3. Introduction to Artificial Intelligence – Eugene Charniak and Drew McDermott
4. Machine Learning – Tom M. Mitchell, McGraw Hill.
5. Deep Learning – Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press.
6. Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow –
7. Aurélien Géron, O'Reilly. AI Superpowers – Kai-Fu Lee

Course code:	Course title: AI Tools and its applications
Course credits: 02 (0:0:2)	Hours/Week: 4
Total contact class: 60	Formative assessment marks: 10
Exam marks: 40	Exam duration: 3 Hours

Course Learning Outcomes (CLOs)

After successful completion of this practical course, students will be able to:

CLO 1: Use AI tools to generate text, images, presentations and reports effectively.

CLO 2: Apply AI tools for solving basic computational and programming problems.

CLO 3: Analyze data and generate meaningful insights using AI assistance.

CLO 4: Compare different AI tools and select appropriate tools for specific tasks.

CLO 5: Demonstrate the use of AI tools in academic work such as resume creation, translation and documentation.

Course Pedagogy

1. Demonstrates the usage of AI tools (e.g., ChatGPT, Canva, Gamma).
2. Hands-On Practical Learning by conducting experiments individually or in small groups.
3. Problem-Based Learning (PBL) - solve real-world or academic problems using AI tools.

List of Practical Experiments:

Part A

1. Create an account and explore features of ChatGPT (text generation, summarization).
2. Use ChatGPT to generate a technical literature review on one of topic (IoT / AI in Healthcare / Smart Grid) and refine it into IEEE abstract format
3. Use Google Gemini to generate a short essay on “AI past and future”.
4. Generate an image of future smart class and analyze
5. Design a project poster using Canva AI tools including smart layout, AI background removal, and presentation enhancement.
6. Create a technical presentation using AI presentation generator on “Future of Edge AI in Engineering Applications”- Gamma.
7. Write a paragraph with mistakes, Upload in Grammarly, Note corrections and Rewrite corrected paragraph
8. Create invitation and flyer for Science day celebration function in your college
9. Develop a no-code, student grievance management prototype using Google AI.

Part B

10. Create AI-generated diagrams for system architecture of a smart campus project using AI diagram tools.
11. Generate an MCQ with 25 questions on Data structure, DBMS and Computer networks with appropriate headings
12. Build a workflow automation using no-code tools for attendance management or report approval system.
13. Compare outputs of two AI chatbots for solving one numerical problem and evaluate response accuracy.
14. Interact with AI Chatbot. Ask 5 academic questions, note responses, Evaluate correctness
15. Use AI tool to record the answers and observation on Operating system
16. Upload erroneous Python/C++ code into AI tool, debug it, optimize it, and explain performance improvements. Create a resume using Gemini
17. Use Microsoft Excel AI functions to clean a raw dataset, detect anomalies, and generate trend analysis with formulas and charts.

Evaluation scheme for practical examination:

Semester: V

Assessment criteria		Marks
Writing	One program from part A	15
	One program from part B	15
Execution	Any one of the written problems	5
Viva voce		5
Total Marks		40

CIE, SEE and QP Pattern for Theory Courses (2 Credits):

Total Lecture hours per paper: 30, No. of Units: 3 (10 Hours Each)

Internal Assessment: C1 - 5 Marks, C2 - 5 Marks

Semester End Theory Exam: C3 - 40 Marks

Question paper pattern for Theory Courses (2 Credits):

Maximum Marks – 40

Part A: Answer any 5 questions out of 6: (5x2=10 Marks) Question No. 1 to 6

Part B: Answer the following questions: (10x3=30 Marks)

7.a) 7. b) or 7. c) 7. d)

8.a) 8. b) or 8. c) 8. d)

9.a) 9. b) or 9. c) 9. d)

CIE, SEE and QP Pattern for Theory Courses (3 Credits):

Total Lecture hours per paper: 44

No. of Units 4 (11 Hours Each)

Internal Assessment: C1 - 10 Marks, C2 - 10 Marks

Semester End Theory Exam: C3 - 80 Marks

Question paper pattern for Theory Courses (3 Credits):

Instructions: Answer Part-A and Part-B:

Part-A

Answer any 10 out of 12 Questions (3 Questions drawn from each unit).

Each question carries 2 Marks. (10 X 2 =20) Q.

No. 1 to Q. No. 12.

Part-B

Answer all the Questions. Each question carries 15 Marks. (4 X 15 =60) (Each question with internal choice and with maximum of 3 sub questions)

Programme Learning Outcomes (PLOs)

PLO 1: Computational Knowledge

Apply the knowledge of computing fundamentals, mathematics, and domain concepts to solve complex computing problems in areas such as operating systems, computer networks, algorithms, artificial intelligence, and data analytics.

PLO 2: Problem solving skills

Identify, formulate, and analyze complex computing problems using logical reasoning, algorithmic thinking, and appropriate computational techniques.

PLO 3: Modern Tool Usage

Select and apply modern tools, technologies, and platforms such as AI tools, data analytics software, block chain systems, and visualization tools for effective problem-solving.

PLO 4: Experimentation & Data Analysis

Conduct experiments, analyze data, and interpret results using statistical, computational, and visualization techniques to support decision-making.

PLO 5: Professional Ethics & Security

Apply ethical principles and commit to professional responsibilities, including data security, privacy, and legal compliance in computing practices.

PLO 6: Team Work

Function effectively as a member or leader in multidisciplinary teams to accomplish shared goals.

PLO 7: Project Management

Apply project management principles, including planning, scheduling, risk management, and resource allocation in software and IT projects.

Sustainable Development Goals (SDGs)

SDG 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5: Achieve gender equality and empower all women and girls.

SDG 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

SDG 13: Take urgent action to combat climate change and its impacts.

Programme Learning Outcomes (PLOs) and Sustainable Development Goals (SDGs) - Course Compliance Matrix

Sl. No.	Course Title	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7
1	Network Security	✓	✓	✓	✓	✓	-	-
2	Network Security Lab	✓	✓	✓	✓	✓	✓	-
3	Design and Analysis of Algorithms	✓	✓	✓	✓	-	✓	-
4	Design and Analysis of Algorithms Lab	✓	✓	✓	✓	-	✓	-
5	Computer Graphics & Multimedia	✓	✓	✓	✓	-	✓	✓
6	Computer Graphics & Multimedia Lab	✓	✓	✓	✓	-	✓	✓
7	Blockchain Technology	✓	✓	✓	✓	✓	-	✓
8	Enterprise Resource Planning	✓	✓	✓	✓	✓	✓	✓
9	Fundamentals of Artificial Intelligence	✓	✓	✓	✓	✓	-	✓
10	AI Tools and its applications	✓	✓	✓	✓	✓	✓	✓
11	Scientific Computing Techniques	✓	✓	-	✓	-	-	-
12	Scientific Computing Techniques Lab	✓	✓	✓	✓	-	✓	-
13	Big Data Analytics	✓	✓	✓	✓	-	-	✓
14	Big Data Analytics Lab	✓	✓	✓	✓	-	✓	✓
15	Web Content Management System	✓	✓	✓	✓	-	✓	✓
16	Web Based Mini Project	✓	✓	✓	✓	✓	✓	✓
17	Software Project Management	✓	✓	-	✓	✓	✓	✓
18	Mobile Communication	✓	✓	-	✓	✓	-	-
19	Power BI	✓	✓	✓	✓	-	✓	-

Sl. No.	Course Title	SDG4	SDG5	SDG9	SDG13
1	Network Security	✓	✓	✓	-
2	Network Security Lab	✓	✓	✓	-
3	Design and Analysis of Algorithms	✓	✓	-	-
4	Design and Analysis of Algorithms Lab	✓	✓	-	-
5	Computer Graphics & Multimedia	✓	✓	✓	-
6	Computer Graphics & Multimedia Lab	✓	✓	✓	-
7	Blockchain Technology	✓	✓	✓	-
8	Enterprise Resource Planning	✓	✓	✓	-
9	Fundamentals of Artificial Intelligence	✓	✓	✓	✓
10	AI Tools and its applications	✓	✓	✓	✓
11	Scientific Computing Techniques	✓	-	✓	-
12	Scientific Computing Techniques Lab	✓	-	✓	-
13	Big Data Analytics	✓	-	✓	✓
14	Big Data Analytics Lab	✓	-	✓	✓
15	Web Content Management System	✓	-	✓	-
16	Web Based Mini Project	✓	✓	✓	-
17	Software Project Management	✓	-	✓	-
18	Mobile Communication	✓	-	✓	-
19	Power BI	✓	-	✓	✓